

Practice Quiz 02C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Conditionals, contrapositives, and free and bound variables.

1. Let G mean *the gate opens* and K mean *the key is accepted*. The statement is $G \rightarrow K$.

- (a) Write its converse and contrapositive as formulas. Make one four-row truth table and say which is equivalent to $G \rightarrow K$.

Solution The converse is $K \rightarrow G$ and the contrapositive is $\neg K \rightarrow \neg G$.

G	K	$G \rightarrow K$	$K \rightarrow G$	$\neg K \rightarrow \neg G$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	T	T	T

The contrapositive is equivalent to the original conditional; the converse is not. ■

- (b) In $\forall s \in \mathbf{R} \exists t \in \mathbf{R} (s + t = c)$, list the free variables and the bound variables.

Solution The variables s and t are bound by their quantifiers. The variable c is free. ■